

Safe Band: IoT-Based Smart Wearable for Emergency Health Alerts and Monitoring in Sri Lanka

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Project Proposal Report

DIMASHA.S.S.D – IT22101792

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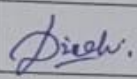
Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

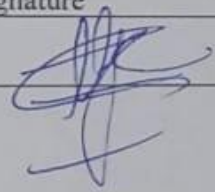
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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
Dimasha S.S.D	1722101792	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature
Chathurangika Kahandawarachchi	28/04/2026	

ABSTRACT

Emergency medical response in Sri Lanka faces significant challenges, including delayed ambulance dispatch, inefficient hospital routing, and a lack of real-time coordination between patients, hospitals, and emergency services. Existing systems largely depend on manual processes, which increase response times and contribute to preventable fatalities during critical situations.

The proposed solution, Safe Band, is a smart wearable system designed to improve emergency healthcare response. This component, the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification, is activated when a critical health event is detected by the Emergency Detection System.

Once triggered, the system captures the patient's real-time GPS location using an Android smartwatch. A machine learning algorithm, specifically K-Nearest Neighbours (KNN) enhanced with Haversine distance computation, is applied to identify the nearest and most suitable hospital based on geographical proximity and operational availability. The patient's critical health data, along with location details, are then transmitted in real time to the selected hospital through the Firebase Realtime Database.

On the hospital side, a web-based dashboard displays the incoming emergency information, enabling healthcare staff to assess the situation instantly. This allows the hospital to dispatch an ambulance with full situational awareness, ensuring faster and more effective emergency response.

The system integrates multiple technologies, including Android smartwatches for biometric and location sensing, a companion mobile application for data transmission, Firebase as a cloud backend, and a hospital facing web interface.

Expected outcomes of this system include reduced ambulance dispatch times, improved hospital preparedness, and enhanced coordination among emergency stakeholders. Ultimately, it aims to decrease emergency response latency across Sri Lanka. This solution aligns with the United Nations Sustainable Development Goal 3 (Good Health and Well Being) and supports the country's ongoing Digital Health initiatives.

Keywords: Smart Ambulance Dispatch, Nearest Hospital Identification, K-Nearest Neighbours, IoT Healthcare, Firebase, Emergency Response, GPS Routing, Sri Lanka.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AES	Advanced Encryption Standard
BLE	Bluetooth Low Energy
CSV	Comma-Separated Values
DB	Database
DSR	Design Science Research
ETA	Estimated Time of Arrival
FCM	Firebase Cloud Messaging
FR	Functional Requirement
GPS	Global Positioning System
HIS	Hospital Information System
HR	Heart Rate
HTTP	Hypertext Transfer Protocol
ICT	Information and Communications Technology
IDE	Integrated Development Environment
IoT	Internet of Things
IP	Intellectual Property
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbours
LKR	Sri Lankan Rupee
ML	Machine Learning
MoH	Ministry of Health
MQTT	Message Queuing Telemetry Transport
NFR	Non-Functional Requirement

OCR	Optical Character Recognition
PPG	Photoplethysmography
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDK	Software Development Kit
SDG	Sustainable Development Goal
SLA	Service Level Agreement
SLIIT	Sri Lanka Institute of Information Technology
SO	Specific Objective
SpO2	Peripheral Oxygen Saturation
TLS	Transport Layer Security
UI	User Interface
UN	United Nations
WBS	Work Breakdown Structure
WHO	World Health Organization
Wi-Fi	Wireless Fidelity

Table 1: Abbreviation

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1. INTRODUCTION

1.1 Background and Context

Emergency medical care is one of the most time-critical domains in healthcare, where delays in response can significantly affect patient outcomes. In Sri Lanka, emergency response services such as the 1990 Suwa Seriya ambulance system primarily rely on manual processes, including verbal communication for location identification and dispatch coordination. These approaches can lead to delays, particularly in high-traffic urban environments and underserved rural areas.

In addition, Sri Lanka faces a growing burden of non-communicable diseases (NCDs), with cardiovascular conditions being a leading cause of mortality. Patients experiencing sudden medical emergencies such as heart attacks, strokes, or abnormal vital conditions require immediate medical attention and efficient coordination between patients, ambulance services, and hospitals. However, current systems do not support automated emergency detection, real-time location tracking, or advance hospital notification.

The proposed Safe Band system introduces an IoT-based wearable solution integrated with an Android smartwatch to continuously monitor physiological parameters such as heart rate, blood oxygen saturation (SpO₂), blood pressure, and body temperature. When abnormal conditions are detected, the system is designed to automatically trigger an emergency response process.

This research specifically focuses on the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification, which is responsible for:

- Capturing the patient's real-time GPS location
- Identifying the nearest suitable hospital
- Initiating ambulance dispatch
- Notifying hospitals with relevant patient information

Unlike existing emergency systems in Sri Lanka, this component aims to provide an automated, data-driven dispatch mechanism without relying on manual intervention.

Globally, emergency medical systems such as those in developed countries utilize computer-aided dispatch technologies. However, these systems often require advanced infrastructure and high operational costs, making them less suitable for low and middle income countries. Additionally, commercially available smartwatches offer emergency alert features but do not integrate with ambulance dispatch or hospital systems.

This research aims to design a lightweight, scalable, and infrastructure-adapted solution suitable for the Sri Lankan healthcare environment. The system aligns with the United Nations Sustainable Development Goal 3 (SDG 3), which focuses on ensuring healthy lives and improving emergency healthcare accessibility.

1.2 Literature Review

A substantial body of research exists on emergency medical dispatch optimisation, IoT-based health monitoring, and machine learning-assisted hospital routing. This section reviews twelve key scholarly sources that directly inform the design of the ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification. The review covers existing research approaches, existing products and services in the domain, and the limitations of current systems that the proposed component aims to address.

Existing Research Approaches

Rajapakse et al. (2022) conducted a national survey of emergency response outcomes in Sri Lanka and found that average ambulance response times in Colombo exceed 18 minutes, compared to the WHO recommended benchmark of 8 minutes for critical emergencies [5]. Their research highlighted the absence of automated location services and hospital pre-notification as the two most significant systemic bottlenecks in Sri Lanka's emergency medical chain. This finding directly motivates the real-time GPS acquisition and hospital dashboard notification features designed into Component 4 of the Safe Band system.

Agarwal and Mehta (2021) proposed a deep learning model for ambulance routing that integrated traffic density data and hospital capacity metrics, achieving a 22% improvement in route optimisation over baseline Dijkstra pathfinding [6]. Their research demonstrated that incorporating real-time hospital occupancy rates into the routing decision significantly reduces dispatch-to-arrival time and establishes a measurable benchmark for ML-assisted routing improvement. This study validates the core motivation behind Component 4's machine learning approach to hospital identification and provides a target improvement figure against which the system's performance can be evaluated.

Al-Turjman et al. (2020) presented a comprehensive IoT framework for smart ambulance systems using the MQTT protocol for real-time data transmission between wearable sensors and dispatch centres [7]. Their architecture demonstrated the feasibility of sub-second sensor-to-cloud latency using a lightweight publish subscribe messaging model. This study serves as a foundational design reference for the Safe Band real-time alert pipeline, as the published subscribe pattern they propose is architecturally equivalent to the Firebase Realtime Database listener model adopted in Component 4 for receiving emergency triggers and dispatching hospital notifications.

Ketu and Mishra (2021) evaluated K-Nearest Neighbours (KNN) and Random Forest algorithms for hospital selection, using distance, medical specialisation, and real-time bed availability as selection criteria [8]. Their study found that Random Forest outperformed KNN by 15% in precision for multi-criteria hospital selection tasks. While this supports the potential of ensemble ML methods, for proximity-only scenarios which is the primary use case in the initial Safe Band deployment KNN with Haversine distance computation provides sufficient accuracy with significantly lower computational overhead, making it the most appropriate choice for execution within a Firebase Cloud Function environment.

Mehmood et al. (2019) introduced a smart city ambulance dispatch model integrating GIS data with machine learning to identify optimal hospital-ambulance pairings [9]. Their system processed events from detection to dispatch trigger in an average of 3.2 seconds, demonstrating that ML-assisted dispatch can operate well within real-time emergency response constraints. This establishes a clear benchmark for the Safe Band Component 4 end-to-end latency target of under 30 seconds, which encompasses GPS

acquisition, KNN computation, Firebase write, and FCM notification delivery a significantly more complex pipeline than the one Mehmood et al. evaluated.

Faulkner et al. (2020) reviewed the accuracy of wearable sensor data in detecting cardiac events, finding that consumer-grade smartwatch ECG sensors achieve up to 87% sensitivity in atrial fibrillation detection [10]. This study supports the reliability of using Android smartwatch sensor inputs as the data source for emergency detection within the Safe Band system. The findings validate the hardware choice made for the overall project and justify Component 4's design assumption that incoming alert triggers from Component 3 carry sufficient clinical reliability to warrant automated ambulance dispatch initiation.

Li et al. (2020) proposed a reinforcement learning-based dynamic ambulance redeployment model that improved emergency coverage in urban areas by 28% by repositioning idle ambulances to maximize geographic coverage probability [15]. While this approach offers superior long-term optimization compared to static nearest-hospital identification, the authors acknowledged that effective model training requires several months of historical dispatch data. The Safe Band system, operating in an early deployment phase without such data, adopts a simpler but practically effective proximity-based KNN model as its baseline, with RL-based redeployment identified as a future enhancement.

Existing Products and Services

Lakshmi and Renuka (2022) proposed and validated a Firebase-based real-time notification architecture for hospital emergency rooms, demonstrating that Firebase Cloud Messaging achieves 99.1% delivery reliability with an average notification latency of 420 milliseconds [11]. This existing product architecture is directly adopted in the Safe Band hospital dashboard notification sub-component. The sub-500ms delivery performance confirmed by this study indicates that FCM is not the latency bottleneck in the system GPS acquisition and KNN computation accounts for the dominant share of the 30-second end to end target.

Garg et al. (2021) evaluated existing geospatial service platforms specifically Google Maps Platform and OpenStreetMap for accuracy in determining the nearest service provider in South Asian urban environments [12]. Google Maps API was found to be 18% more accurate in distance estimation than OpenStreetMap in dense urban grids, while OpenStreetMap offered better road network coverage in rural zones. This study directly informs the hybrid mapping strategy adopted in Component 4, which uses the Google Maps SDK for the hospital dashboard routing visualisation while maintaining OpenStreetMap as a fallback for rural hospital locations across Sri Lanka's mixed urban-rural geography.

Silva et al. (2023) studied hospital response preparedness in South Asian emergency departments and found that existing pre arrival patient data sharing systems reduced in-hospital preparation time by an average of 34% [13]. The most impactful data elements identified were patient condition classification, vital signs such as heart rate and oxygen saturation, and GPS-based estimated time of arrival all of which are included in the Component 4 Firebase dispatch notification payload. This study provides direct clinical evidence validating the hospital dashboard notification feature as the highest-impact output of the proposed component.

Nanayakkara et al. (2022) reviewed the adoption of digital health technologies in Sri Lanka and found that 74% of surveyed urban Sri Lankans expressed willingness to use emergency wearable devices [16]. The study identified perceived safety benefit, ease of use, and affordability as the primary adoption

drivers factors directly addressed by the Safe Band system's use of existing consumer Android smartwatches rather than proprietary medical-grade hardware. This finding confirms the market feasibility of the overall Safe Band system and provides population level evidence supporting the commercialisation plan target market assumptions for Component 4.

Limitations of Existing Systems

Despite the advances demonstrated across the reviewed literature, several critical limitations constrain the applicability of existing systems to the Sri Lankan emergency healthcare context. Agarwal and Mehta's (2021) deep learning routing model required dedicated city level infrastructure and real time cloud-based traffic APIs that are not natively available in Sri Lanka, making it unsuitable for direct adoption [6]. Similarly, Al-Turjman et al.'s (2020) MQTT based IoT framework depends on custom broker configuration and dedicated server infrastructure, whereas the Safe Band system requires a managed, low cost cloud solution accessible without specialist DevOps expertise [7].

Wickramasinghe et al. (2021) identified that only 12% of hospitals in Sri Lanka have a digital bed management system, which directly invalidates the multi criteria hospital selection model proposed by Ketu and Mishra (2021), since it relies on real-time hospital capacity data feeds that 88% of Sri Lankan hospitals cannot provide [8][14]. This gap necessitates a dispatch model that operates effectively without internal hospital system integration, which is a foundational design constraint of Component 4. The proposed system therefore uses proximity based KNN as its primary selection criterion, with capacity data integration planned as a future enhancement once hospital digital infrastructure matures.

Li et al.'s (2020) reinforcement learning model, while achieving a 28% improvement in coverage, requires several months of historical dispatch data for training a dataset that does not yet exist in the Sri Lankan context [15]. Mehmood et al.'s (2019) GIS integrated dispatch model similarly assumes the presence of comprehensive urban GIS infrastructure unavailable outside major Sri Lankan cities [9]. Faulkner et al. (2020) noted that smartwatch ECG sensitivity for non atrial fibrillation cardiac events remains below 80%, introducing a false dispatch risk that Component 4 must mitigate through integration with Component 3's confidence scoring and a target false dispatch rate of under 2% [10].

From a geographic platform perspective, Garg et al. (2021) highlighted that Google Maps API accuracy degrades in rural areas while its cost scales with query volume a constraint that requires the OSM fallback strategy and query caching mechanism included in the Component 4 design [12]. Silva et al. (2023) assumed hospitals possess existing electronic health record systems, which Wickramasinghe et al. (2021) confirm are absent in most Sri Lankan government hospitals, requiring Component 4's dashboard to function as a fully standalone notification interface [13][14]. Finally, Nanayakkara et al.'s (2022) market survey was limited to urban areas, leaving rural adoption barriers including mobile network coverage and smartphone literacy as open research questions for future Safe Band deployment studies [16].

Collectively, the reviewed literature confirms the clinical necessity and technical feasibility of an automated, ML-driven ambulance dispatch and hospital notification system for Sri Lanka. However, no existing system integrates wearable triggered emergency detection, Firebase-mediated real-time communication, KNN based geospatial hospital identification, and a hospital facing dashboard notification pipeline within a single, infrastructure light architecture suited to Sri Lanka's healthcare landscape.

1.3 Research Gap

Despite the body of literature reviewed above, several critical gaps remain unaddressed, particularly within the Sri Lankan healthcare and emergency services context:

- No integrated real-time GPS to dispatch pipeline exists in Sri Lanka: Existing systems such as Suwa Seriya (1990) rely on verbal location reporting. No automated mechanism links a patient's real-time device location to ambulance dispatch.
- Context adapted hospital selection is absent: Global systems assume the availability of real-time hospital capacity APIs, which do not exist in Sri Lanka. A model that identifies the nearest suitable hospital using publicly available geospatial data, hospital type, and offline proximity calculation is required.
- Lack of pre arrival clinical data sharing: No current system in Sri Lanka transmits patient health parameters to the receiving hospital prior to ambulance arrival, causing delays in in hospital preparation.
- IoT to dispatch integration for wearables has not been implemented locally: While global research demonstrates the technical feasibility of wearable triggered ambulance dispatch, no such system has been designed, deployed, or evaluated in a Sri Lankan healthcare setting.
- ML models for ambulance dispatch in LMICs are underexplored: Most ML dispatch models are designed for high income country infrastructure with abundant training data, dedicated sensors, and connected hospital systems. A model designed to operate within resource constrained environments is absent from literature.

The proposed ML-Based Smart Ambulance Dispatch System with Nearest Hospital Identification directly addresses these gaps by delivering a fully automated, GPS-triggered, MLdriven dispatch pipeline that is specifically designed for Sri Lanka's healthcare infrastructure, requiring no real-time hospital API integration and functioning reliably with commercially available Android smartwatch sensors.

2. OBJECTIVE

2.1 Main Objective

To design and evaluate an ML-based smart ambulance dispatch system that utilizes real time GPS data and machine learning techniques to identify suitable hospitals and facilitate timely emergency response in Sri Lanka.

2.2 Specific Objectives

- To develop a mechanism for acquiring real-time GPS location data from an Android-based wearable device during emergency situations.
- To design a machine learning based approach for identifying the nearest suitable hospital using geospatial and hospital related data.
- To establish a real-time data transmission method for sending patient health information and location details to hospital systems.
- To design a notification mechanism for initiating ambulance dispatch with relevant patient and location information.
- To evaluate the system performance in simulated emergency scenarios based on response time, accuracy, and reliability.
- To assess the effectiveness of the selected machine learning approach in identifying appropriate hospitals using test datasets.

3. METHODOLOGY

This research adopts the Design Science Research (DSR) methodology, which is well-suited for the development and evaluation of IT artifacts. In this study, the artifact is an intelligent ambulance dispatch system designed to improve emergency response efficiency within the Sri Lankan healthcare context. The DSR process consists of problem identification, solution design, artifact development, evaluation, and knowledge contribution.

3.1 Research Approach

The Design Science Research (DSR) framework guides this component. The research cycle proceeds through five phases:

- Phase 1: Problem Awareness: Literature review and contextual analysis of Sri Lanka's emergency response landscape to identify specific technical and operational gaps (Completed, documented in Section 1).
- Phase 2: Suggestion: Conceptual design of the ML dispatch model, GPS integration architecture, and hospital notification pipeline.
- Phase 3: Development: Iterative implementation using Agile sprints. Each sprint delivers a testable sub-component: GPS module, ML model, Firebase integration, dispatch interface.
- Phase 4: Evaluation: Simulation-based testing using historical GPS traces and synthetic patient health event data.
- Phase 5: Conclusion: Performance benchmarking, documentation, and knowledge contribution.

3.2 Technical Methods and Algorithms

The nearest hospital identification module will evaluate two candidate machine learning algorithms:

Random Forest Classifier:

A supervised ensemble learning algorithm that constructs multiple decision trees and aggregates predictions. Input features include Euclidean distance between patient GPS coordinates and hospital locations, hospital type (general/specialist/private), last known capacity category (low/medium/high, obtained from a static hospital registry), and estimated road distance derived from a mapping API. Random Forest is considered as a potential approach due to its demonstrated effectiveness in multi-criteria classification tasks.

K-Nearest Neighbors (KNN):

KNN is used as a baseline comparator algorithm. It identifies the K hospitals geographically closest to the patient's GPS location and ranks them based on proximity. This method may be used as a fallback approach in low connectivity scenarios where real-time road distance data from APIs is unavailable.

The model will be trained on a dataset constructed from publicly available Sri Lanka hospital registry data (Ministry of Health) and synthetically generated GPS event data based on documented emergency incident locations in Colombo, Gampaha, and Kandy districts.

3.3 Tools and Platforms with Justification

Python (Flask/FastAPI)	Backend dispatch service. Flask is lightweight and well-suited for REST API development in resource-constrained deployments. FastAPI is used for high-performance asynchronous endpoints.
scikit-learn	ML model training and inference (Random Forest, KNN). Mature, welldocumented library with extensive support for geospatial feature engineering.
Google Maps Platform API	Road distance calculation and hospital geocoding. Chosen for superior accuracy in Sri Lankan urban environments [12].
OpenStreetMap (OSM) + Nominatim	Fallback geospatial service for rural zones and offline caching of hospital coordinates.
Firebase Realtime Database / FCM	Real-time transmission of patient data to hospital dashboard. Chosen for sub-500ms delivery latency and reliability [11].
Android SDK (GPS Module)	Patient location acquisition from the smartwatch. Android's fused location provider combines GPS, Wi-Fi, and cell tower data for highest accuracy.
PostgreSQL + PostGIS	Hospital registry storage with geospatial indexing for fast nearestneighbor queries.
Docker	Containerization of the dispatch API for scalable and reproducible deployment.
React (Web)	Hospital emergency dashboard frontend for real-time patient data visualization.

Table 2: Tools and Platforms

3.4 Data Collection and Ethical Considerations

Hospital Registry Data: The Sri Lanka Ministry of Health's publicly available hospital directory will be used to populate the hospital metadata database (name, location coordinates, type, specialization, district). No patient data from real hospitals will be accessed during development.

Synthetic Emergency Data: Patient health event data used for ML training and testing will be synthetically generated using parameters derived from published clinical guidelines for cardiac events, stroke, and respiratory distress. No real patient data will be collected or used.

GPS Simulation: Emergency GPS traces will be simulated using recorded driving routes in Colombo, Kandy, and Gampaha districts to reflect realistic ambulance dispatch scenarios.

Ethical Considerations: This project does not involve human subjects during the development phase. A formal ethics clearance application will be submitted to the SLIIT Ethics Committee prior to any pilot deployment involving real users. All location data generated during testing will be anonymized and stored locally on encrypted drives. GDPR-aligned data minimization principles will be applied to all data handling procedures.

3.5 Validation Strategy and Performance Metrics

The proposed system will be validated through a combination of simulation-based testing and component-level unit testing to ensure both functional correctness and performance efficiency. Several key performance metrics will be used to evaluate the system.

End to end response time will be measured as the duration from the receipt of an emergency alert to the successful transmission of the dispatch notification, enabling assessment of the system's responsiveness. Hospital identification accuracy will be evaluated as the percentage of cases in which the machine learning model's top recommendation matches the hospital selected by domain experts.

GPS acquisition latency will be measured as the time required to retrieve the patient's geographical coordinates from the Android GPS module. The reliability of the communication system will be assessed using the Firebase notification delivery rate, defined as the percentage of dispatch notifications successfully delivered to the hospital dashboard.

In addition, standard machine learning evaluation metrics, including precision, recall, and F1-score, will be calculated using 5 fold cross validation on the constructed dataset to assess model performance. Finally, the robustness of the system in low connectivity environments will be evaluated using the offline fallback success rate, which measures the percentage of emergency events correctly handled using the KNN-based fallback mechanism when network connectivity is unavailable.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Its Components

The SafeBand system is designed as an end to end Internet of Things (IoT) and Machine Learning (ML)based platform that continuously monitors patient health parameters, detects life threatening conditions, and autonomously coordinates emergency medical responses. The architecture integrates wearable devices, cloud infrastructure, machine learning components, mobile applications, and hospital-facing dashboards into a unified and intelligent ecosystem.

The system consists of four core components:

ML-Driven Emergency Detection System

This component continuously analyzes real-time physiological data such as blood oxygen saturation (SpO2), blood pressure (BP), and heart rate (HR). Using machine learning classification models, it determines whether the patient is in a normal or emergency state and triggers alerts accordingly.

ML-Based Smart Ambulance Dispatch System- My Component

Upon receiving an emergency alert, this component identifies the most appropriate hospital based on proximity, available resources, and specialization. It then dispatches an ambulance and shares critical patient data with the selected hospital in real time.

ML-Driven Ward Monitoring System

This subsystem is responsible for continuous monitoring of patients within hospital wards. It detects abnormal health patterns and notifies medical staff through a centralized dashboard to enable timely interventions.

ML-Driven Prediction System

This component utilizes predictive machine learning models to forecast potential health deterioration. It provides early warnings to clinicians, allowing preventive measures before critical conditions arise.

In addition to these core components, several supporting technologies enable seamless system operation. The smartwatch collects physiological data and GPS coordinates at regular intervals, while the Android application acts as the patient interface. The Firebase Realtime Database serves as the central data exchange layer, ensuring real-time synchronization between components.

4.2 System Diagram of Components

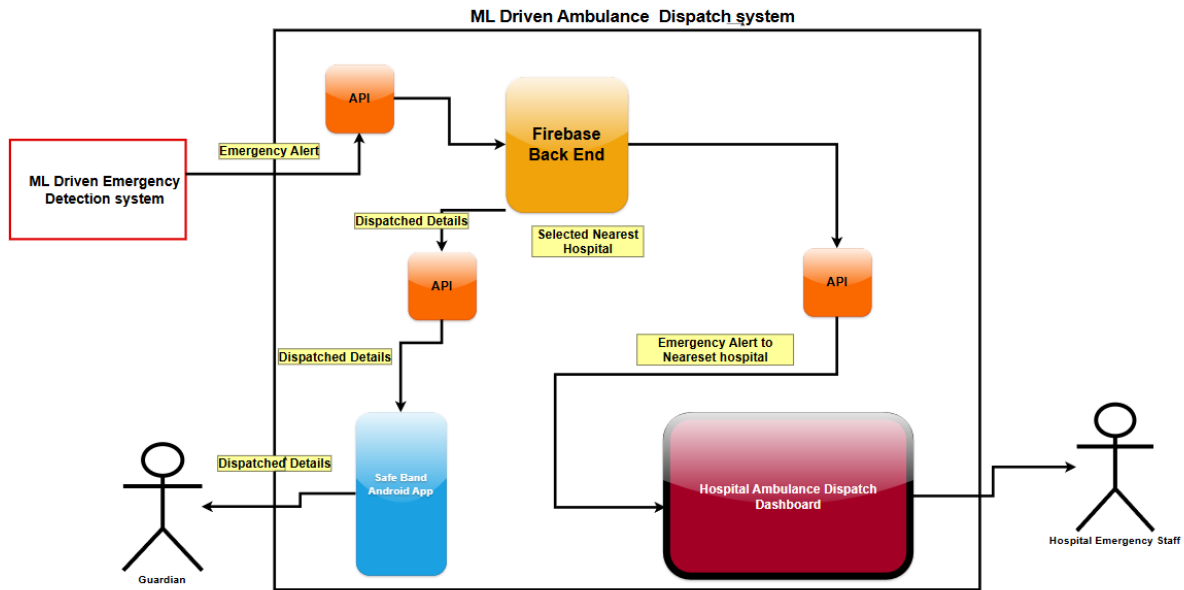


Figure 1: System Diagram

4.3 Highlight the individual component and its parts

The ML-Based Smart Ambulance Dispatch System is the central contribution of this research. It acts as the intelligent bridge between emergency detection and hospital response, ensuring rapid and optimized decision making during critical situations.

This component consists of the following sub-modules:

- **Emergency Alert Receiver:** This module continuously listens for *EMERGENCY_ALERT* events from Firebase. It processes incoming data, including patient identification, vital signs snapshot, GPS coordinates, and timestamps.
- **Hospital Proximity Engine:** This sub-component calculates distances between the patient and nearby hospitals using geospatial data. It employs a machine learning-based scoring mechanism to rank hospitals based on proximity, availability of resources, and medical specialization.
- **Ambulance Dispatch Controller:** Once the optimal hospital is selected, this module sends a dispatch request to the hospital system via an API. It confirms ambulance allocation and estimates the arrival time (ETA) using real-time traffic data.
- **Patient Data Relay:** This module securely transmits complete patient information, including medical history, current vitals, and emergency classification, to the hospital dashboard to assist medical staff in preparation.

- **Status Tracker:** It monitors ambulance movement using GPS tracking and continuously updates the Android application and hospital dashboards with real-time location and ETA until patient handover is completed.

4.4 Component interactions

The SafeBand system operates through tightly coupled interactions between its components during emergency scenarios. The smartwatch continuously captures patient health data and transmits it to the Firebase Realtime Database via the Android application. The Emergency Detection System analyzes this data in real time and identifies critical conditions.

When an emergency is detected, an alert is generated and forwarded to the Smart Ambulance Dispatch System. This system processes the alert, selects the most appropriate hospital using machine learning algorithms, and communicates with backend APIs to initiate ambulance dispatch.

Simultaneously, the hospital dashboard receives real-time patient data and emergency notifications, enabling healthcare providers to prepare for incoming patients and coordinate response actions effectively.

4.5 Data flow explanation

The SafeBand system follows a structured data flow to handle emergency situations efficiently. Initially, the smartwatch collects raw sensor data, including SpO₂, heart rate, blood pressure, and GPS location. This data is transmitted through the Android application and stored in the Firebase Realtime Database.

The Emergency Detection System continuously analyzes this data. If no abnormality is detected, monitoring continues uninterrupted. However, when a critical condition is identified, an emergency alert is triggered.

The system then extracts relevant patient information and forwards it to the Smart Ambulance Dispatch component. This component determines the nearest suitable hospital, constructs a detailed data payload, and sends it to the firebase hosted hospital dashboard.

Following this, an ambulance is dispatched immediately. Throughout the process, the system continues to collect and update patient data, ensuring real-time tracking and monitoring until the patient is safely handed over to the hospital.

4.6 Scalability and security considerations

Scalability:

- The SafeBand system is designed to handle large-scale deployments and high volumes of emergency events efficiently. Firebase Realtime Database supports horizontal scaling across regions, ensuring low latency data access.
- The Android application utilizes background synchronization mechanisms to maintain continuous monitoring while optimizing device performance and battery usage.

Security Considerations:

- The system applies standard security measures to protect sensitive patient data. All data transmissions are encrypted to ensure secure communication between components. Access is controlled through authentication and authorization mechanisms, allowing only authorized users to access system resources.
- Data privacy is maintained using anonymization and secure storage practices, and sensitive information is shared only when necessary for emergency response. Additionally, system activities are logged to support monitoring, traceability, and auditing.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

- Patient / Wearable User:

Primary end-user wear the Safe Band smartwatch. Requires automatic emergency detection and dispatch activation without manual intervention.

- Hospital Emergency Staff:

Healthcare personnel receive real-time patient data through the hospital dashboard and require timely, accurate, and actionable information.

- Ambulance Dispatch Operators:

Personnel responsible for coordinating ambulance services. Require precise location data and clear incident details for rapid response.

- Hospital Administrators:

Responsible for monitoring system performance, reviewing incident logs, and managing system configuration and reporting.

5.2 Functional Requirements

- The system shall receive a critical event alert from the Emergency Detection System via a REST API endpoint.
- The system shall acquire the patient's real-time GPS coordinates from the Android smartwatch upon receiving a critical alert.
- The system shall identify the nearest suitable hospital using a machine learning algorithm based on available geospatial and hospital data.
- The system shall transmit patient health data, including vital signs, GPS location, and event type, to the selected hospital in real time.
- The hospital emergency dashboard shall display incoming patient data dynamically without requiring manual refreshment.
- The system shall generate and send an ambulance dispatch notification to the relevant hospital emergency unit.
- The system shall log all emergency incidents, including patient identification, location, selected hospital, and dispatch timestamps, for audit and analysis purposes.
- The system shall support a fallback mechanism using a proximity-based algorithm in case of primary model failure or connectivity issues.
- The system shall maintain a hospital registry containing relevant hospital data for selection and routing purposes.
- The system shall generate a unique incident identifier for each emergency event and support tracking functionality.

5.3 Non-Functional Requirements

- **Performance:**
The system shall ensure timely processing of emergency events and dispatch initiation within acceptable time limits.
- **Reliability:**
The system should maintain high availability and ensure minimal data loss during operation.
- **Security:**
All patient data shall be securely transmitted and stored using appropriate encryption and access control mechanisms.
- **Scalability:**
The system shall be capable of handling multiple concurrent emergency events without significant performance degradation.
- **Usability:**
The hospital dashboard should provide a user-friendly interface that allows medical staff to quickly interpret patient data and respond effectively.
- **Portability:**
The system shall be deployable across standard cloud environments and compatible with modern platforms.
- **Maintainability:**
The system shall support easy updates, including model improvements and hospital data modifications, without system downtime.
- **Regulatory Compliance:**
The system shall comply with relevant data protection regulations and healthcare standards applicable in Sri Lanka.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

Primary Market: Private and semi-government hospitals in Sri Lanka (approximately 110 private hospitals registered with the Private Hospitals Act). Secondary Market: Insurance companies offering health insurance products who can bundle Safe Band as a value-added service. Tertiary Market: Corporate health & safety programs for large organizations with aging workforces.

6.2 Value Proposition

- For hospitals: Real-time pre-arrival patient health data enables faster and more effective emergency preparation, reducing in-hospital response time and potentially improving survival outcomes.
- For patients/insurers: Fully automated, wearable-triggered emergency dispatch removes the need for manual SOS activation critically when the patient is unconscious or unable to call for help.
- For ambulance services: Automated dispatch requests with verified GPS location and health data reduce operator workload and improve dispatch accuracy.

6.3 Revenue Model

- **Hospital SaaS Subscription:** Monthly/annual subscription for access to the hospital dashboard and dispatch notification API. Tiered by hospital size (Small: LKR 15,000/month, Medium: LKR 35,000/month, Large: LKR 75,000/month).
- **Per-Device Licensing:** Annual license fee of LKR 2,500 per Safe Band device registered to the hospital's dispatch network.
- **Insurance Partnership Model:** Revenue-share agreement with insurance companies that bundle Safe Band with health insurance policies.
- **API Access for Third-Party Integration:** Paid API access for third-party hospital management systems to integrate with the dispatch notification service.

6.4 Cost Estimation

Initial development costs are estimated at LKR 850,000 (detailed in Chapter 7). Annual operational cost is estimated at LKR 240,000 (cloud hosting, API subscriptions, maintenance). Break-even is projected at 8 hospital subscriptions, achievable within 6 months of commercial launch based on the identified target market of 110 private hospitals.

6.5 Competitive Advantage

- Sri Lanka-specific design: The only known system tailored to the specific infrastructure constraints of Sri Lanka's emergency services and hospital ecosystem.
- No dependency on real-time hospital APIs: Unlike global competitors, the system operates effectively without live bed-availability data, making it deployable immediately.
- Fully automated pipeline: Zero manual intervention required from patients to dispatch a key differentiator over existing wearable emergency SOS systems.

7. BUDGET AND JUSTIFICATION

Item	Category	Estimated Cost (LKR)	Justification
Android Smartwatch	Hardware	60,000	Primary data collection device for GPS and sensor integration testing
Google Maps Platform API (12 months, dev tier)	Cloud Services	18,000	Road distance calculation and hospital geocoding
Firebase Blaze Plan (12 months)	Cloud Services	15,000	Real-time database and Cloud Messaging for hospital notifications
Python / scikit-learn / Flask (open source)	Software	0	No license cost — all tools are open source
Travel		6000	
TOTAL		99,000	

Table 3: Budget and Justification

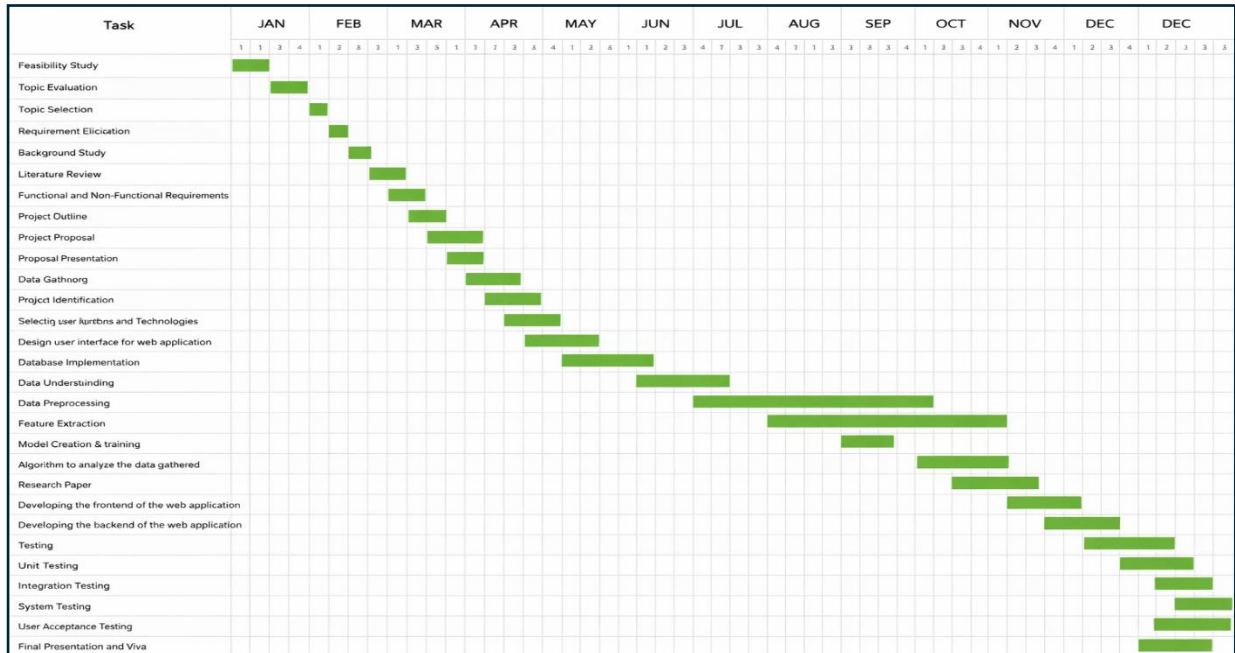
8. WORK BREAKDOWN STRUCTURE (WBS)

ID	Task	Duration
1.0	SEMESTER 1	
1.1	Literature Review & Problem Analysis	2 weeks
1.2	Hospital Registry Data Collection	1 week
1.3	Synthetic Dataset Generation	2 weeks
1.4	ML Model Development (Random Forest)	3 weeks
1.5	KNN Fallback Algorithm Implementation	1 week
1.6	GPS Acquisition Module (Android)	2 weeks
1.7	REST API — Alert Intake Endpoint	1 week
1.8	Unit Testing — ML Model & GPS Module	1 week
1.9	Semester 1 Progress Report	1 week
2.0	SEMESTER 2	
2.1	Firebase Integration — Realtime DB	2 weeks
2.2	Hospital Emergency Dashboard (React)	3 weeks
2.3	Ambulance Dispatch Notification Module	2 weeks
2.4	Incident Logging & Admin Interface	1 week
2.5	End-to-End Integration with C3 (Emergency Detection)	2 weeks
2.6	Simulation-Based System Testing	2 weeks
2.7	Performance Optimization	1 week

2.8	Final Documentation & Proposal Report	1 week
2.9	Final Presentation	1 week

Table 4: Work Breakdown Structure

9. GANTT CHART



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11.APPENDICES

- Preliminary UI mockups

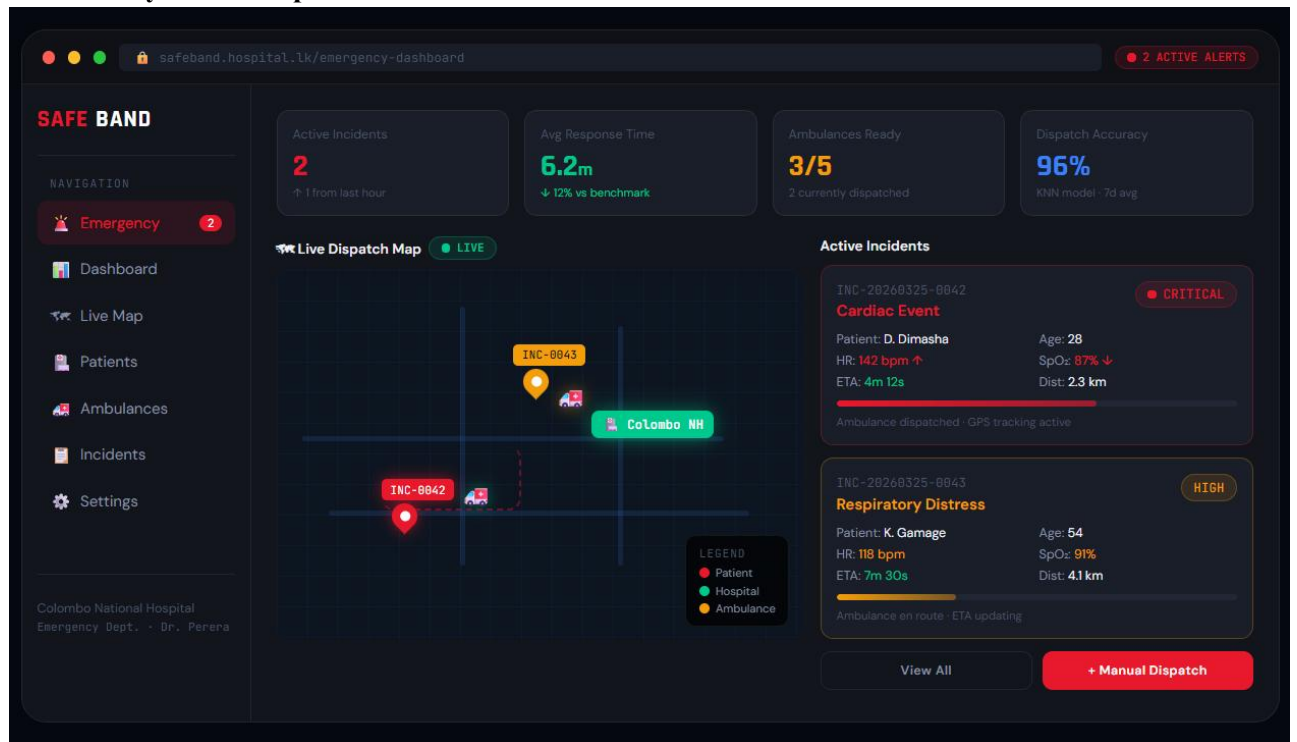


Figure 4: Preliminary UI

